MILITARY SPECIFICATION

INTEGRATOR, BALL AND DISC

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope.- This specification describes the requirements for the procurement of two types of Ball and Disc Integrator, BUORD drawing 1675234 and BUORD drawing 1981275. The Ball and Disc Integrators are precision mechanical devices having two angular inputs and one angular output with the angular output equal to 0.4 times the integral of one angular input with respect to the other. Ball and Disc Integrator BUORD drawing 1675324 has the output shaft and the lead screw input shaft on the same side of the assembly, while Ball and Disc Integrator BUORD number 1981275 has them on opposite sides of the assembly.

1.2 Classification.- The Ball and Disc Integrators shall be of the following types:

Type I	BUORD Drawing No. 1675234,	Integrator, Ball and Disc
Type II	BUORD Drawing No, 1981275,	Integrator, Ball and Disc

2. APPLICABLE DOCUMENTS

2,1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner

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FSC 1290

PPP-B-636	Boxes, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical. and Associated Equipment, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specifications for Design and Procurement of

STANDARDS

MILITARY

Mil-sTd-129	Marking of Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUORD dWg 1675234	Integrator, Ball and Disc
BUORD Dwg 1981275	Integrator, Ball and Disc
BUORD SA-253&03	Assembly Ball Disc Integrator T.F.
BUORD SA-2530407	Assembly Ball Disc Integrator T.F.

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: AMERICAN TRUCKING ASSOCIATION , INC PUBLICATION

Motor Freight Classification Rules

(Application for copies **should be addressed to the Issuing Officer, American Trucking Association, Inc.,** 1424 16th **Street**, N.W., Washington **6, D.C.)**

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production samples.- Unless otherwise specified in contract or order, pre-production samples of the Ball and Disc Integrator shall be manufactured BY the methods and procedures proposed for the production lot. These samples shall be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the 'requirements of this specification and the design. , See 4.1.2.

3.1.2 Materials.- Unless otherwise specified in the contract, re**quisition, or order, all materials and purchased parts entering into the final assembly of) the Ball and Disc Integrator shall conform to the detailed requirements given on BUORD drawing 1675234 or** 1981275, and in Specification MIL-F-18870.

3.1.3 Finishes, coatings, construction, and dimensions .- Finishes, coatings,'- construction, and dimensions for the Ball and Disc Integrator shall be in accordance with the applicable drawings, specifications, and publications listed in BUORD **drawing l65234 or** 1981275.

3.1.4 Moisture, fungus, and corrosion.- The Ball and Disc Integrator shall be rendered moisture, fungus, and corrosion resistant in at-' cordance with detail' requirements specified on BUORD drawing 1675234 or 19$1275,

3,1.5 Marking.-" Marking of assemblies, sub-assemblies, and parts shall be in accordance with BUORD drawing 1675234 or 1981275.

3.1.6 Interchangeability.- Component parts of the assembled units' (unless otherwise specified on the applicable drawings) fabricated in accordance with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. '" If,during assembly of the fabricated parts of a unit, selection or fitting ie. required, then changes in dimensions are necessary,' and revision and correction of the applicable drawings is required. See 3.1.10.

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PUBLICATIONS

BUREAU OF NAVY WEAPONS

NAVWEPS OD 12351	Description and operating Instruc - t ions for Ball-Disc Integrator Test Fixture (Type A) Shop Acces - . sory SA 2530403
NAVWEPS OD 17185	Inscription and Operating Instruc - tions for Ball-Disc Integrator Test Fixture (Type B) Shop Accessory SA 2530407-
NAVORD OS 6341	Miscellaneous General Ordnance , ,. &sign Requirements
WR-1l	Design and Test of Packaging, Packing Shipping and Handling Equipment for . Weapon System Components .

(Copies of specifications, standards, drawings, and publica- tions requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as . directed by the contracting officer.)

:.:2.2 Other publications. - The following documents form **a part of this specification.** Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL **BUREAU OF STANDARDS PUBLICATIONS**

Handbook H28	Screw-thread Standards for Federal . Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freiht Classification Rules
t.,

(Application for copies should be addressed to the Consolidated . o. Classification Committee, 202 Chicago Union Station, Chicago 6,Ill)

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3.1.7 Gages. The **contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final. Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as maybe required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.**

3.1.8 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on-the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.9 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the procuring activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially **temporary in** character, and usually represents a departure limited to a specific period; to a specific number of items, or to the contract to which it applies. **It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.**

3.1.10 Conflicting requirements. - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - Definitions applicable to this specification and associated. drawings are as follows:

3.2.1 Breakaway torque. - Breakaway torque is defined as the minimum torque necessary to start rotation of the disc input shaft.

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3.2.2 Zero position. The zero position of the lead screw shaft of the Ball-and Disc Integrator is that angular position for which no disc shaft angular velocity should be transmitted to the output shaft, and from which lead screw shaft angle is measured.

3.2.3 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which presents an-opportunity for fungi growth.

3.3 performance requirements and product characteristics. - The Ball and Disc Integrator shall meet the following performance requirements and product characteristics:

3.3.1 Alignment. - When tested in accordance with 4.4.4, with the lead screw shaft at the zero position and the disc input-shaft rotated at the rate of 300 revolutions per minute for 600 plus 60 minus zero revolutions, the output shaft shall not rotate more than 0.02 revolution.

3.3.2 **Accuracy. -** When tested in accordance with 4.4.4, and for the specific test conditions listed below, the accuracy of the Ball and Disc Integrator shall be **such that none of the errors in output shaft angle exceeds 3.7 revolutions for either direction of lead screw angular displacement from the zero position.**

Lead Screw Shaft Input (Revolutions)	**Disc Shaft* Input (Revolutions)**	**output (Revolutions)**
1	1000	400
3	1000	1200
5	1000	2000

* at a rate of 300 revolutions per minute.

3.4 Environmenmtal requirements. - The Ball and **Disc** Integrator **shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.**

3.4.1 Vibration, non-opepating condition. - The Ball and Disc Integrator shall withstand the Type I vibration requirements specified in Standard MIL-STD-167, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Vibration-transportation. - The Ball and Disc Integrator **after being** subjected to the tests specified in 4.5.2 shall show no damage and shall meet all the requirements of Section 3.

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3.4.3 Shock, non-operating condition. - The Ball and Disc Integrator shall withstand the Type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.4 Shock-transportation. - The Ball and Disc Integrator after being subjected to the tests specified in 4.5.4 shall show no damage and shall meet all the requirements of Section *3.*

3.4.5 Temperature, operating condition. - The Ball and Disc Integrator shall withstand operational ambient temperatures of between zero degrees Fahrenheit (minus 18 degrees centigrade) and 185 degrees Fahrenheit *(85* **degrees** centigrade) in accordance with Specification MIL-F-*18870,* shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification.

3.4.6 Temperature, non-operating condition. - The Ball and Disc Integrator shall withstand storage temperatures of between minus 40 degrees Fahrenheit (minus 40 degrees centigrade) and 16o degrees Fahrenheit (71 degrees centigrade) in accordance with Condition II of OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4,7 Humidity. - The Ball and Disc Integrator shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.8 Tropical environment. - The Ball and Disc Integrator shall withstand the tropical environmental conditions specified in Specification MIL-E-5272 under "Fungus Resistance Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification.

3.4.9 Operating life. - After 2500 hours of operation in accordance with *4.5.9,* the output shaft angle of the Ball and Disc Integrator shall be 2400 plus or minus 3.7 revolutions for a lead screw shaft input of 6 turns in either direction from the zero position and a disc shaft input of 1000 revolutions at 300 revolutions per minute.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification: Particular attention shall be paid to marking of parts and assemblies, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government . The Government reserves the right to perform any of the inspection set forth in this specification **where such inspect** ions are deemed necessary to ensure SUP- **plies** and services conform to prescribed requirments.

4.1.1 sampling. - Unless otherwise specified by the contract or purchase order, and when applicable, the sampling plans and procedures used in the determinant ion of the acceptability of products submitted by " a supplier shall be in accordance with 4.1.2 and 4.1.3.

4.1.2 Pre-production sample. - Unless otherwise specified in the contract or order (see 6.2), two Bell and Disc Integrators manufactured in accordance with 3.1.1 shall be furnished to the activity designated **by the procuring agency see** 6.2) for testing in accordance with 4.5. Further production of the item, prior to completion of the tests and approval of the sample by the procuring agency, shall be at the contractors risk.

4.1.3 **Periodic production sample. - Periodic production samples shall. be randomly selected as follows from production units which have satisfactorily passed the tests of 4.4, and subjected to the tests of** 4.6 by the activity designated in the contract or order (see 6.2):

Step 1. Upon initiation of production select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty- **five (25) units.**

Step 3. Upon satisfactory compliance of each unit from three (3)-consecutive groups, select one (1) unit from each group of fifty (50) units **. Continue this procedure until a defective unit is found.**

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Step 4. When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to Step 1 and proceed as indicated.

4.2 classification of tests. - The inspection and testing of the Ball and Disc Integrator shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each **Ball and Disc** Integrator submitted for acceptance under contract. These teats" are detailed in 4.4.

4.2.2 Pre-production and periodic production tests.. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.2 and 4.1.3 which are representative of the production of the Ball and Disc Integrator after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the procuring activity.

4.2.3 Tests after delivery, - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests. specified herein, as deemed necessary, after installation of the Ball and-" Disc Integrator. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of thee contract.

4.3 Test equipment. - The following items of test equipment are required to' perform the acceptance tests set forth in this specification.

4.3.1 Shop accessory SA-2530403, Ball-Disc Integrator Test Fixture (Type A)

4.3.2. Shop accessory SA-2530407, Ball-Disc Integrator Test Fixture " (Type B).

4.4 Acceptance tests.

4.4.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4.2.1 to assure that the Ball and Disc Integrator is in compliance with the requirements Of 3.1, 3.3, 3.5 and 3.6. Manufacturer's certification maybe used in lieu of material tests, provided that such certificates contain actual test results or other verifiable, inspection data.

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4.4.2 Test conditions, -Unless otherwise specified, the Ball and Disc Integrator shall be subjected to acceptance tests under the following conditions:

4.4.2.1 **Temperature** : Room ambient 65 to 95 degrees Fahrenheit (18 to 35 **degrees** centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2. 3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified,~sha114be~of sufficient accuracy and quality to assure the adequacy and accuracy of test results relative to required tests.

4.4.4 Test procedure. - The Ball and Disc Integrator shall be tested in accordance with the detailed test procedure given in OD l2351 or OD 17185 to **determine compliance with the requirements of** *3.1, 3.3, 3.5,* and 3 . 6 .

4.4.4.l 'Preparation for shipment. - The preservation, packaging, and packing of the units shall be examined to determine compliance with the requirements of section 5.

4.5 Pre-production and periodic production sample tests.'- Ball and'" Disc Integrators to these tests shall be considered not suitable for service use, and shall not be issued to the fleet, nor used in equipment . . to be issued to the fleet.

4.5.1 Vibration, non-operating condition. - The Ball and Disc Integrator, rigidly mounted in a suitable vibration fixture, shall be subjected, to vibration tests so as to meet the requirements specified in 3.4.1, and shall meet all the performance requirements and product characteristics of this specification after the vibration tests.

4.5.2 Vibration, transportation. - The Ball and Disc Integrator, packaged and packed as defined in section 5, shall **be vibration tested in accordance with procedures specified in Weapons Requirement WR 1l.**

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4.5.3 Shock, non-operating condition. - The Ball and Disc Integrator, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in *3.4.3*, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.4 Shock, transportation, storage, and handling. - The Ball and Disc Integrator, packed and packaged as defined in section 5, shall be subjected to the appropriate shock isolation tests specified in Weapons Requirement WR 11.

4.5.5 Temperature, operating condition. - The Ball and Disc Inte - grater shall be subjected to temperature tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.6 Temperature, non-operating condition. - The Ball and Disc Integrator shall be subjected to temperature tests so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.7 Humidity. - The Ball and Disc Integrator shall be subjected to humidity tests so as to meet the requirements specified in 3.4.7, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.8 Tropical environment. - The Ball and Disc Integrator shall be exposed to fungus tests so as to meet the requirements specified in 3.4.8, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.9 Operating life. - The Ball and Disc Integrator shall be operated for 2500 hours under the environmental conditions specified in 4.4.2 with a disk shaft speed of 300 revolutions per minute, the lead screw shaft advanced from 6 turns off the zero position in either direction to 6 turns off the zero position in the opposite direction at a shaft speed of 0.0048 revolution per hour, and an output torque of 0.15 plus or minus 0.05 ounce-inch. At the end of the 2500-hour operating period, angular output of the Ball and Disc Integrator shall be noted for a lead screw shaft input of 6 turns in either direction from the zero position and a disk shaft input of 1000 revolutions at 300 revolutions per minute to determine compliance with the requirements of 3.4.9.

4.6 Failure of the Ball and Disc Integrator to meet any of the requirements and tests of this specification shall be considered cause for rejection

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4.7 **Waiver** of tests. Tests specified in 4.5.2 and 4.5.4 may be **waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.**

4.8 Functional tests. - At the conclusion of the tests specified in 4.5.2 and 4.5.4, functional and other tests, as applicable, shall be conducted on the Ball and Disc Integrator to determine freedom from operational malfunction and compliance with all the requirements of section so

5. **PREPARATION FOR DElIveRY**

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging."- Unit packaging of the Ball and Disc Integratorr shall be in accordance with Method IIe of Specification MIL-P-116. The unit package shall contain one Ball and Disc Integrator.

5.1.1.3 Intermediate **package. - Not applicable.**

5.1.1.4 Cushioning. - **The unit package, and pack, shall Incorporate sufficient cushioning material; bracing, or other adequate shock absorbing devices, to ensure that the Ball and Disc Integrator will meet all the requirements of section 3 after being tested in accordance with 4.5.2 and 4.5.4.**

At **the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.**

5.1.2 **LevelB. - Not** applicable.

5.1.3 Level c.

5.1.3.1 cleaning, drying and preservative applications - Not applicable.

5.1.3.2 Unit *Packaging.* Unit packaging of the Ball and Disc Integrator shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one Ball and Disc Integrator.

5.1,3.3 Intermediate package. - Not applicable.

5.2 Packing. .

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5.2.1 Level A.

5.2.1.1 Exterior COntainers. *Ball* and Disc Integrator packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, class 2 (weight not to exceed 150 lbs) or in a fiberboard box in accordance with Specification PPP-B-636, class 2 (weight not to exceed 70 lbs).

5.2.2 Level B. - Not applicable.

5.2.3 Level C. 'The Ball and Disc Integrater Packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5•3 Marking,

5.3.1 Special markings. - None, unless otherwise specified.

5.3.2 Normal markings. - In **addition to the markings required by contract or order, unit packages, and shipping containers shall be marked** in accordance with the requirements of MIL-STD-129, **including serial numhers .**

6. NOTES

6.1 Intended use. The Ball and Disc Integrators intended for use in the Angle Solvers Mark **18 Mod 2 and** Mod *3,* **the Fire Control System** Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-l, -2, *-3,* -4, or other suitable application. -

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity required including samples required for pre-production and periodic tests (see 4.5).
Sampling if different from 4.1.1
d. If pre-production or periodic samples are not required (see 3.1,1).
e. Activity to conduct Dre-production and periodic tests (see *4.1.2* **and** 4.1.3).
f. Materials, if different from 3.1.2.
g. Level of "preparation for delivery" (see section 5).
h. Additional marking if required (see 5.3.1).

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6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to paragraph 4.7 of this specification.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For level **A.** - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions **for** redistribution anywhere.

For level B. This level shall be used only when it is-definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For level C. -This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any Ball and Disck Integrator which fails to meet the requirements of this specification within one year after delivery **because of a defect in material or workmanship** or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

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SPECIFICATI ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-I-22357(Wep) INYEGRATOR, BALL AND DISC

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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